

Nathan Simon

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EDUCATION & HONORS

The New School, Eugene Lang College, B.A. Psychology, New York, NY | 2022 - Winter 2025

- GPA: 3.96/4.00, Summa Cum Laude, Dean's Honor List
- Relevant Coursework: Machine Learning, Audio Signal Processing, Cognitive Psychology, Research Methods, Statistics

EXPERIENCE

Hyfe, AI Workflow Intern | Spring 2026 - Present

Healthtech startup specializing in private, on-device cough monitoring for pharmaceutical research

- Built AI tooling that translated acoustic model results into clear, interactable summaries for non-technical researchers and clinicians
- Designed and shipped Slack-native AI agents adopted across Hyfe's finance, sales, and ops functions, replacing recurring back-office work in QuickBooks, Wise, HubSpot, and Google Workspace
- Built an AI pipeline that consolidated years of Slack, email, and Google Drive history into a unified database of every clinical study Hyfe has run, giving the team a single source of truth for prior research

Human Research and Development Lab, Technical Researcher | Winter 2025 - Spring 2026

Academic research lab (led by Dr. Ruthe Foushee) studying pre-linguistic baby-caregiver communication through computational audio analysis.

- Led development of machine learning models for audio event classification in baby-caregiver dyads, training on a proprietary human-labeled audio dataset
- Managed expert-level human data labelers to create proprietary, high-quality audio datasets of pre-linguistic baby-caregiver dyad communication

Inner Compass Initiative, Communications Intern | Summer 2025

Nonprofit raising awareness about psychiatric drug withdrawal and informed consent in mental health care.

- Supported awareness campaigns on psychiatric drug withdrawal and assisted staff interfacing with the NYC Department of Homeless Services
- Served as principal researcher for Cooper Davis, Director and CEO of Inner Compass

Global Health Strategies, Communications and AI Intern | Summer 2024

GHS is a global consulting firm that provides strategic services, advocacy, policy analysis, and communications.

- Implemented a firm-wide AI adoption plan for US-based offices, including tool evaluation, workflow integration, and staff training
- Supported strategic communications for the Alzheimer's Disease Data Initiative and other Alzheimer's programs in the private office of Bill Gates

Mount Sinai Suicide Research & Prevention Lab, Lab Assistant | Academic Year 2023

- Identified potential participants at multiple psychiatric wards for ongoing studies
- Trained in administering cognitive tests for study onboarding

Proprietary Investing, Self-Directed | 2024 - Present

- Generated +232% total return over 507 trading days (Feb 2024 - Feb 2026) on personal capital across equities, options contracts and shorts
- 1.75 annualized Sharpe (levered), +47% CAPM alpha, 1.48 beta to S&P 500; trailing 200-day Sharpe of 3.33

SELECTED PROJECTS

Embodied Communication | Psychology, ML

- Built human-in-the-loop ML pipeline classifying caregiver-infant interactional events from daylong audio of 68 Tzeltal Maya dyads using Template Matching, MFCC features and Wavelet Scattering Networks
- Achieved 85% top-1 accuracy on soothing and infant vocalization events; surfaced novel finding that breastfeeding functions as a regulatory mechanism around sleep
- Co-author on in-progress manuscript (code to be open-sourced); presented at Spring 2026 conferences

[Next-Gen Constitutional Classifier Replication](#) | Mechanistic Interpretability

- Replicated and extended Anthropic's Constitutional Classifiers++ on Gemma 3; ~10M-param linear probe on frozen activations + single step classifier beats ShieldGemma 27B on ToxicChat (F1 0.762 vs 0.729) at 1000x fewer params
- Combined text classifier + activation probe detects 67.4% of WildJailbreak attacks at 5% FPR

[Bloomberg Terminal Agent](#) | Equity Analysis, Agentic Computer Use

- Built an Agentic AI interface for Bloomberg terminals, using Proprietary APIs and exposed endpoints on the machine, that can retrieve and analyze data in Excel, as well as export data for model analysis